



Faculty of Engineering and Technology
Electrical and Computer Engineering Department
ENCS4130 – Computer Network Laboratory

Experiment No. 09

Cisco ASA Firewall Configuration

■ Objectives

- (a) Learn how to configure a Cisco adaptive security appliance (ASA) firewall.
- (b) Learn how to set up network address translation (NAT).
- (c) Learn how to configure a named access control list (ACL).

■ Requirements

- (a) Cisco Packet Tracer software.
- (b) Two ASA firewalls, two routers, and four switches.
- (c) One wireless access point and four servers (two web and two DNS servers).
- (d) Two PCs, one laptop, one tablet, one smartphone, and one printer.

■ Pre-Lab Task

Students are required to build the network topology shown in Figure 3, including all notes, as outlined in Section 9.2.1. In addition, they must complete the configurations for all routers (OSPF and BGP routing) and all servers, as detailed in Sections 9.2.2 and 9.2.3.

9.1 Introduction

A firewall is a network security device that *inspects data packets* and determines whether to allow or block them according to configured security rules. These rules can be based on several factors, including source and destination IP addresses, port numbers, and protocol types. Its primary function is to *create a protective barrier* between a *trusted internal network* and *untrusted external networks*.

By enforcing these controls, firewalls help prevent unauthorized access and protect against various threats such as hackers, bots, and malware attempting to infiltrate or overwhelm a private network. Without this safeguard, malicious actors could freely exploit the system and compromise sensitive information.

Firewalls can be implemented as either software or hardware (or a combination of both) and come in various forms, including:

- **Packet-Filtering Firewalls:** These analyze packets of data being sent to or from a network and allow or block them based on a set of rules, such as IP addresses, ports, or protocols.
- **Stateful Inspection Firewalls:** These track the state of active connections and make decisions based on the context of the traffic, not just individual packets.
- **Proxy Firewalls:** These act as intermediaries between the internal network and the Internet, making requests on behalf of users and then forwarding the responses back to them.
- **Next-Generation Firewalls:** These offer more advanced features, such as application awareness, intrusion prevention systems, and deep packet inspection, to provide more comprehensive security.
- **Cloud Firewalls:** These are hosted in the cloud and are typically used to protect cloud-based infrastructure and services.

9.1.1 Cisco Adaptive Security Appliance (ASA)

The Cisco ASA is a versatile network security device that integrates multiple security functions into a single platform. These include a *firewall*, *antivirus*, *intrusion prevention*, and *virtual private network (VPN)* capabilities. In this lab, the ASA will be employed to demonstrate how such a device can provide **proactive threat defense** by stopping unauthorized access before threats can spread within the network.

By default, the Cisco ASA *blocks all inbound traffic* from external networks, creating a secure perimeter around the internal network. Despite this strict default posture, the ASA is equipped with *advanced traffic inspection capabilities* that allow it to recognize and permit legitimate connections based on customizable security policies. These policies can be tailored to allow specific types of traffic while maintaining strong security controls.

A) Demilitarized Zone (DMZ)

A key feature of the ASA firewall is its ability to create a DMZ, as illustrated in Figure 1. A DMZ is a logically or physically isolated subnetwork positioned between an organization's internal network and external, untrusted networks such as the Internet. Services

that require external access, such as domain name system (DNS), file transfer protocol (FTP) servers, web servers, and proxy servers, are typically placed within the DMZ. This setup allows for secure interaction with external networks while safeguarding the internal local area network (LAN). Essentially, the DMZ acts as a buffer zone, enabling controlled access to public-facing services while preventing direct exposure of internal systems.

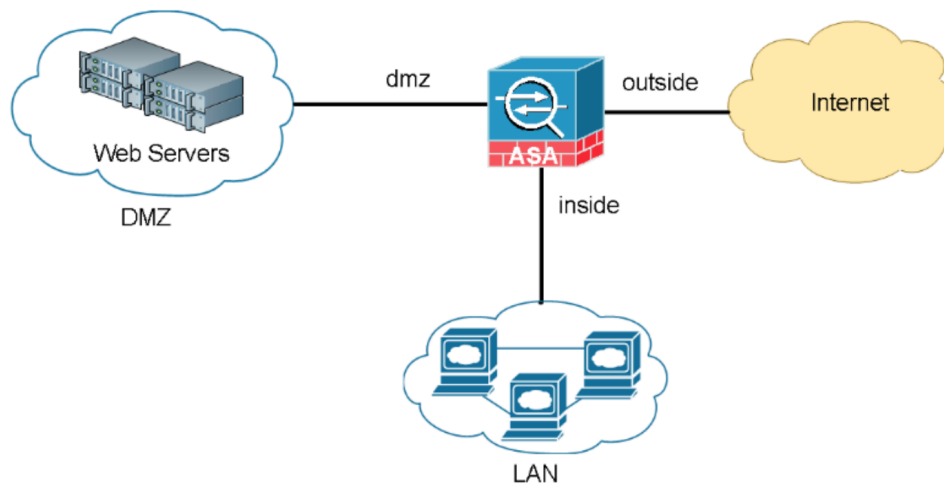


Figure 1: Organizational network overview featuring Cisco ASA as the security gateway.

B) Network Address Translation (NAT)

NAT is a key feature of the Cisco ASA firewall that enables secure and efficient communication between internal (private) and external (public) networks. NAT works by **translating private IP addresses into public ones**, as illustrated in Figure 2. This process hides internal IP addresses from external users, helping to *prevent direct attacks on internal devices* and *preserve the limited pool of public IP addresses*.

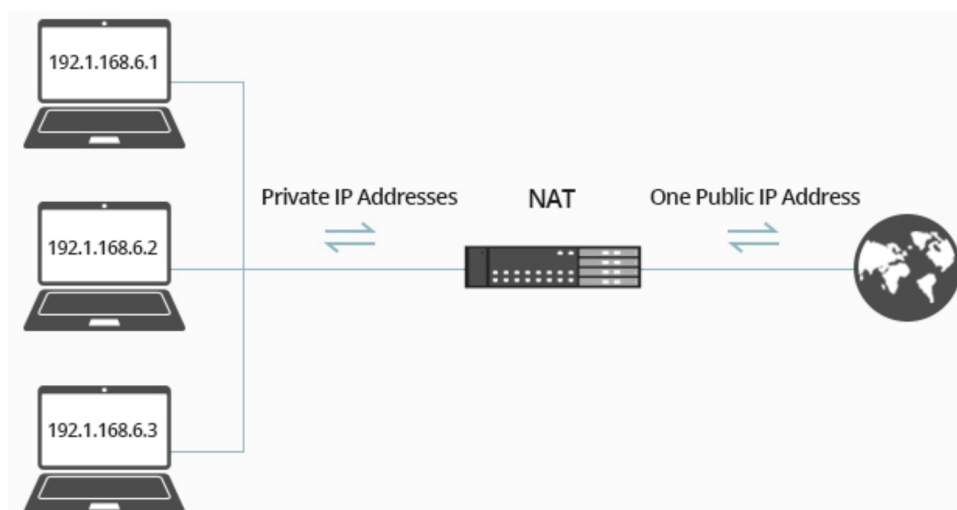


Figure 2: Overview of NAT.

Cisco ASA supports multiple NAT types, each suited to different networking needs:

1. **Static NAT:** Maps a single private IP address to a single public IP address. This is

commonly used for devices such as web and DNS servers that must be consistently reachable from the Internet.

2. **Dynamic NAT:** Maps private IP addresses to public IP addresses from a pre-defined pool. It is useful when multiple internal hosts need occasional external access.
3. **Port Address Translation (PAT)** (also known as NAT Overload): Maps multiple private IP addresses to a single public IP address using unique port numbers for each session. This is the most efficient type of NAT and is widely used to conserve public IPs.

C) Configuration of Cisco ASA Firewall

Configuring a Cisco ASA firewall typically involves defining network interfaces, assigning security levels, NAT, and applying access control policies. The following is a general structure for setting up a Cisco ASA firewall with DMZ and NAT using the standard command syntax.

1. Configure Interfaces and Security Zones

Assign a logical name, security level, and IP address to the ASA interface, and enable it.

```
ciscoasa(config)# interface <interface-name>
ciscoasa(config-if)# nameif <zone-name>
ciscoasa(config-if)# security-level <security-level>
ciscoasa(config-if)# ip address <ip-address>
                           <subnet-mask>
ciscoasa(config-if)# no shutdown
```

- **<interface-name>**: The hardware interface identifier (e.g., Fa0/0, Gig1/2).
- **<zone-name>**: Logical name for the interface (e.g., inside, outside, dmz). This simplifies configuration by allowing you to reference the interface by name instead of its physical identifier.
- **<security-level>**: A numeric value from 0 to 100 that represents the trust level of the interface—the higher the value, the more trusted the network. Typically, **inside** = 100 (most trusted), **dmz** = 50 (moderately trusted), and **outside** = 0 (least trusted).
- **<ip-address>** and **<subnet-mask>**: IP configuration for the interface.

2. Configure PAT

Translate internal (private) IPs to a public IP.

```
ciscoasa(config)# object network <object-name>
ciscoasa(config-network-object)# subnet <private-ip>
                                   <subnet-mask>
ciscoasa(config-network-object)# nat (<in-if>, <out-if>)
                                   dynamic interface
```

- **<object-name>**: A descriptive name for the network object (e.g., **INTERNAL-PAT**). This name simplifies configuration by allowing you to reference the object by name—rather than repeating full IP addresses or subnet definitions—across different settings such as NAT and access control lists (ACLs).
- **<private-ip>** and **<subnet-mask>**: Internal network IP address and mask.
- **<in-if>**: Interface name for the internal zone (e.g., **inside**).
- **<out-if>**: Interface name for the external zone (e.g., **outside**).

3. Configure Static NAT

Map a specific internal host to a fixed public IP address.

```
ciscoasa(config)# object network <object-name>
ciscoasa(config-network-object)# host <private-ip>
ciscoasa(config-network-object)# nat (<in-if>, <out-if>)
                                static <public-ip>
```

- **<private-ip>**: Internal host IP address.
- **<public-ip>**: Assigned public IP address for external access.

4. Enabling Legitimate Traffic Flow from Lower to Higher Security Levels

In a Cisco ASA firewall, *traffic from a higher to a lower security level* (e.g., **inside** → **outside**) *is allowed by default*. However, to allow traffic from a lower to a higher security level, such as external access to web and DNS services in the DMZ from the Internet (i.e., **outside** → **dmz**), you must explicitly permit the traffic using ACLs.

Use the **access-list** command to specify the type of traffic you want to allow:

```
ciscoasa(config)# access-list <acl-name> extended permit
                                <protocol> <src-addr> <dst-addr>
```

- **<acl-name>**: Name of the access control list (e.g., **OUTSIDE-ACCESS**).
- **<protocol>**: Protocol to permit (e.g., **tcp**, **udp**, **icmp**, or **ip** for all).
- **<src-addr>**: Source IP address or network.
- **<dst-addr>**: Destination IP address or network.

Then, use the **access-group** command to bind the ACL to an interface in a specific direction:

```
ciscoasa(config)# access-group <acl-name> <in|out>
                                interface <interface-name>
```

- **<in|out>**: Direction of traffic relative to the interface.
- **<interface-name>**: Interface where the ACL is applied (e.g., **outside**).

The ASA firewall permits traffic initiated from the internal LAN to the DMZ or the Internet by default, since the LAN has a higher security level. However, if this traffic is a request—such as an HTTP request or a DNS query—the corresponding

response (e.g., HTTP response or DNS reply) would be blocked by default, as it originates from a lower security level returning to a higher one.

Fortunately, the ASA firewall is **stateful**, meaning it can track and allow return traffic if it *inspects* the connection. By default, the ASA inspects some protocols like **FTP**, **DNS**, and **TFTP** using a **global policy map** named `global_policy` and a **class map** named `inspection_default`. To enable inspection of additional protocols such as **HTTP**:

```
ciscoasa(config)# policy-map global_policy
ciscoasa(config-pmap)# class inspection_default
ciscoasa(config-pmap-c)# inspect http
```

This configuration ensures that return traffic (e.g., HTTP responses) is allowed, maintaining secure and functional two-way communication.

5. Configure Static Route

Define a default route or a route to a specific destination network to ensure that the ASA firewall knows how to forward traffic destined for external networks.

```
ciscoasa(config)# route <interface-name> <dst-network>
                        <subnet-mask> <next-hop-ip-address>
```

- `<interface-name>`: Outbound interface for the route (e.g., `outside`).
- `<dst-network>`: Destination network address (e.g., `0.0.0.0` for default route).
- `<subnet-mask>`: Mask for the destination (e.g., `0.0.0.0` for default).
- `<next-hop-ip-address>`: Next-hop router's IP address.

6. Configure DHCP

In Cisco ASA, the DHCP server is **interface-specific**, meaning it must be explicitly enabled on each interface where IP address assignment is required. The DHCP service allows the ASA to automatically assign IP addresses and related configuration to devices within a designated network zone.

```
ciscoasa(config)# dhcpd address <start-ip>-<end-ip>
                        <interface-name>
ciscoasa(config)# dhcpd dns <dns-server-ip>
ciscoasa(config)# dhcpd enable <interface-name>
```

- `<start-ip>-<end-ip>`: Specifies the range of IP addresses to be assigned to DHCP clients (e.g., `172.17.X.3 - 172.17.X.20`).
- `<dns-server-ip>`: DNS server IP to be provided to DHCP clients.
- `<interface-name>`: Logical name of the interface on which the DHCP service is enabled (e.g., `inside`, `dmz`).

Note:

The IP address assigned to `<interface-name>` will automatically be used as the **default gateway** for DHCP clients on that network segment.

9.2 Procedure

In this lab, we will configure Cisco ASA 5506-X firewalls for two completely separate networks: Birzeit (AS100) and Gaza Education (AS200). The Birzeit Network includes an internal LAN where hosts obtain IP addresses automatically through DHCP, as well as a DMZ hosting public servers with static IP addresses. NAT and dynamic PAT enable internal hosts to access the internet securely, while firewall rules protect the internal network and regulate access to the DMZ.

The Gaza Education Network follows a similar structure, with an internal LAN using static IP addresses and a DMZ containing public servers with static IPs. Its ASA firewall also provides NAT and dynamic PAT to ensure secure outbound access, while enforcing security policies that protect both the internal network and the DMZ.

Within each network, routing is handled by open shortest path first (OSPF), while the border gateway protocol (BGP) is used to exchange routes between Birzeit and Gaza. This configuration keeps the two networks logically separate while ensuring secure and efficient connectivity to the internet.

9.2.1 Building the Topology

Construct the network topology shown in Figure 3, which consists of the following devices:

- Firewalls: From “*Network Devices - Security*”, use **5506-X**.
- Routers: Use **Router-PT**.
- Switches: Use **Switch-PT**.
- Access point: From “*Network Devices - Wireless Devices*”, use **AccessPoint-PT**.
- Servers, PCs, Laptop, Tablet, Smartphone, and Printer: From “*End Devices*”, use **Server-PT**, **PC-PT**, **TabletPC-PT**, **SMARTPHONE-PT**, and **Printer-PT**.
- Connections: Use the “**Automatically Choose Connection Type**” option to connect firewalls, routers, switches, and end devices.

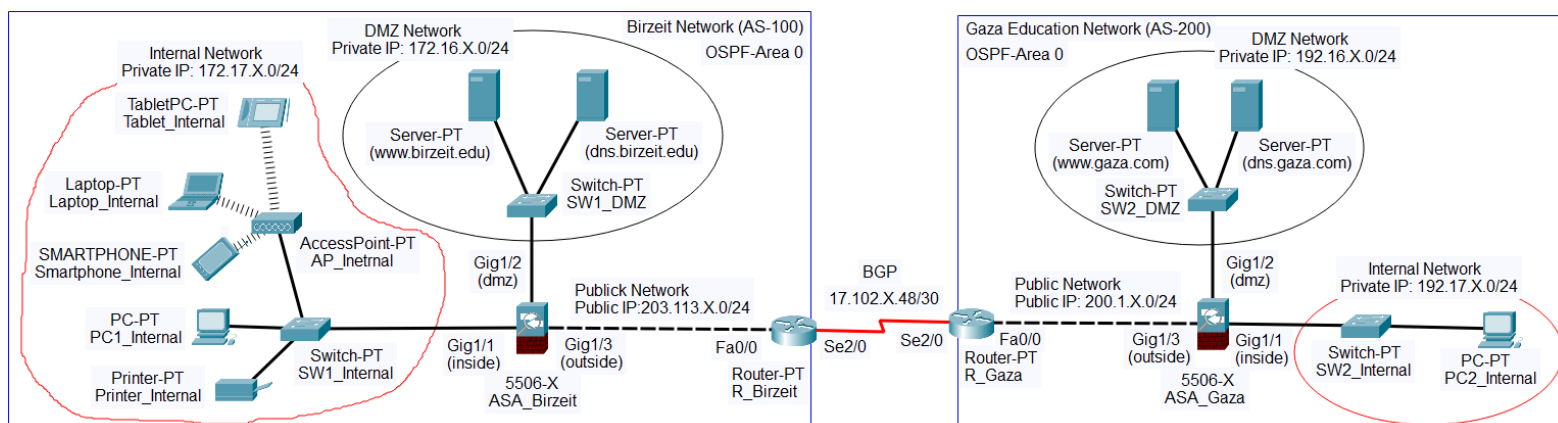


Figure 3: Firewall topology.

9.2.2 Routers Configuration

A) Configuring the Routers' Interfaces

Set up the interfaces on the **R_Birzeit** router using the following commands:

```
R_Birzeit(config)# interface FastEthernet0/0
R_Birzeit(config-if)# ip address 203.113.X.2 255.255.255.0
R_Birzeit(config-if)# no shutdown
```

```
R_Birzeit(config)# interface Serial2/0
R_Birzeit(config-if)# ip address 17.102.X.49 255.255.255.252
R_Birzeit(config-if)# no shutdown
```

Configure the **R_Gaza** router similarly, using its designated IP addresses.

B) Configuring OSPF and BGP Routing Protocols

On **R_Birzeit**, configure OSPF with process ID 1 to advertise the public network and redistribute BGP routes into OSPF:

```
R_Birzeit(config)# router ospf 1
R_Birzeit(config-router)# network 203.113.X.0 0.0.0.255
                        area 0
R_Birzeit(config-router)# redistribute bgp 100 subnets
```

Next, configure BGP by specifying the eBGP neighbor (**R_Gaza**) and redistributing OSPF routes into BGP:

```
R_Birzeit(config)# router bgp 100
R_Birzeit(config-router)# neighbor 17.102.X.50 remote-as 200
R_Birzeit(config-router)# redistribute ospf 1
```

Repeat the same configuration steps on **R_Gaza**, adjusting for its public network, AS number, and the appropriate neighbor IP.

9.2.3 Servers Configuration

Configure the Web and DNS servers for both the Birzeit and Gaza Education Networks as described in **Experiment No. 6**.

A) Birzeit Web Server (www.birzeit.edu)

- Assign a static IP configuration, as shown in Figure 4.
- Enable HTTP and HTTPS (Secure HTTP) protocols.
- Customize the `index.html` page to display the title: “**Birzeit University**”.

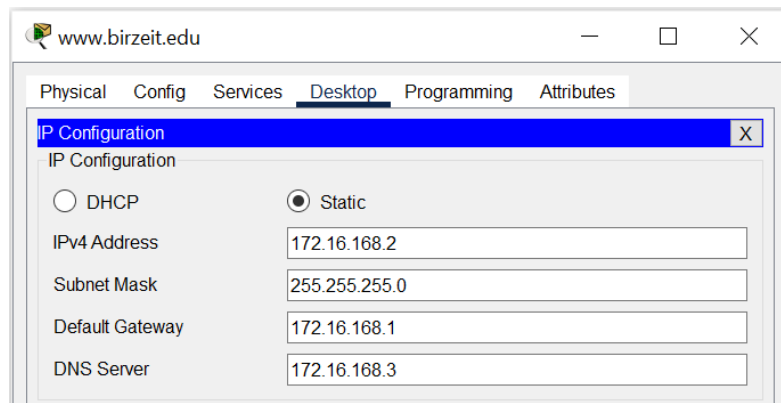


Figure 4: IP configuration of Birzeit Web server.

B) Birzeit DNS Server (dns.birzeit.edu)

- Assign a static IP configuration.
- Enable only the DNS service.
- Add the resource records listed in Table 1, as illustrated in Figure 5.

Table 1: Resource records in Birzeit DNS server.

Type	Name	Value
A	www.birzeit.edu	The IP address of the www.birzeit.edu server
NS	gaza.com	dns.gaza.com
A	dns.gaza.com	The public IP address of the dns.gaza.com server

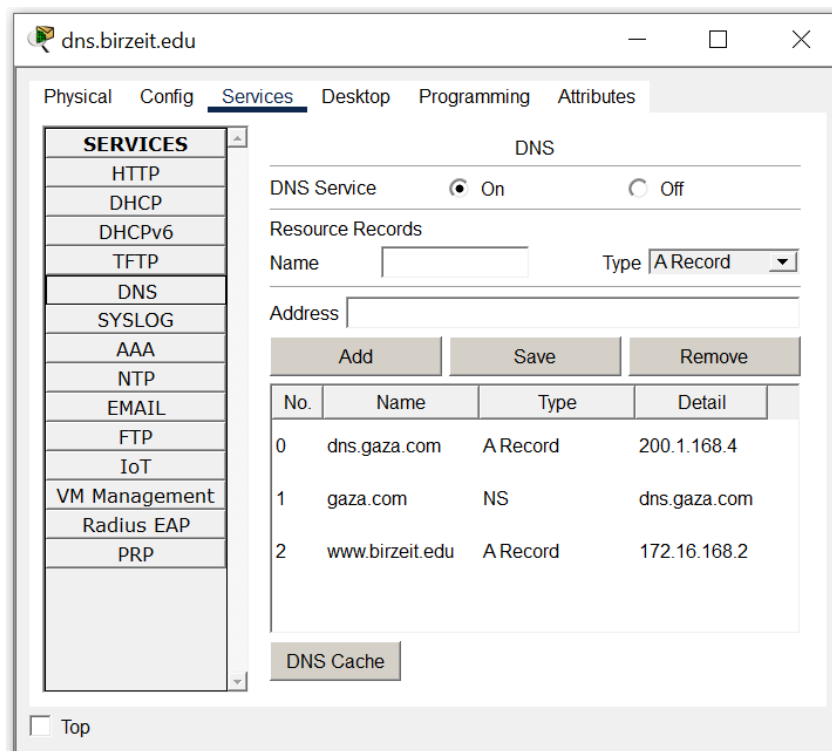


Figure 5: DNS service configuration of Birzeit DNS server.

C) Gaza Web Server (www.gaza.com)

- Assign a static IP configuration.
- Enable HTTP and HTTPS (Secure HTTP) protocols.
- Customize the `index.html` page to display the title: “**Gaza Education Network**”.

D) Gaza DNS Server (dns.gaza.com)

- Assign a static IP configuration.
- Enable only the DNS service.
- Add the resource records listed in Table 2, as illustrated in Figure 6.

Table 2: Resource records in Gaza DNS server.

Type	Name	Value
A	www.gaza.com	The IP address of the www.gaza.com server
NS	birzeit.edu	dns.birzeit.edu
A	dns.birzeit.edu	The public IP address of the dns.birzeit.edu server

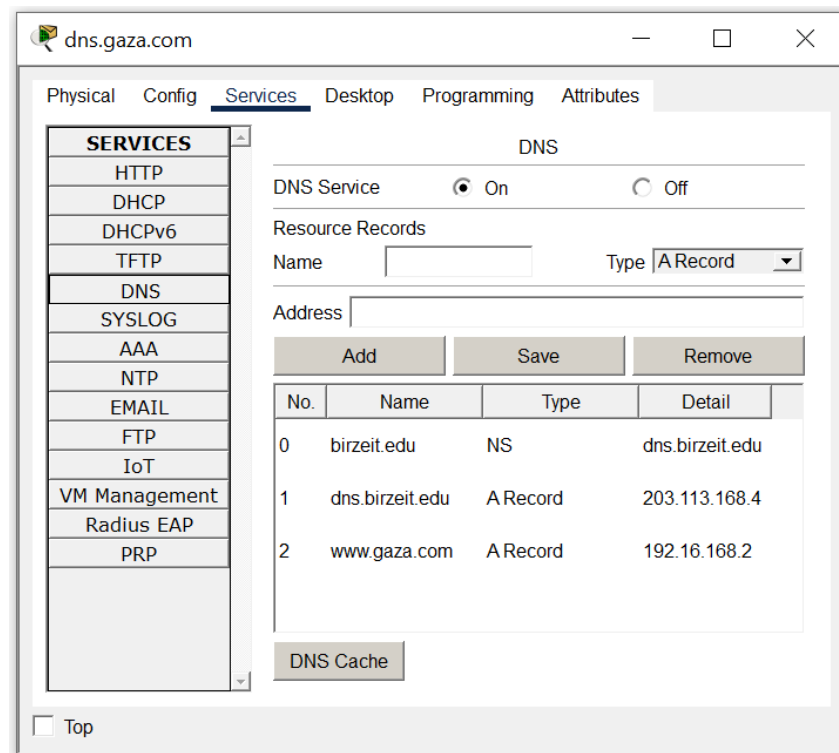


Figure 6: DNS service configuration of Gaza DNS server.

9.2.4 ASA Firewalls Configuration

In this section, we configure the firewalls for both networks. The Birzeit and Gaza ASAs require the same general setup—interfaces, security zones, PAT and NAT rules, access

lists, traffic inspection, and default routing. To illustrate the process clearly, the following steps present the configuration on **ASA_Birzeit** as an example.

A) Configuring Privileged Mode Password

The default password for the privileged mode on the ASA firewall is the **blank**. So, when requested for the privileged mode password, just press the **Enter** key. The following commands change the password to “**encs4130**”:

```
ASA_Birzeit> enable
Password:
ASA_Birzeit# configure terminal
ASA_Birzeit(config)# enable password encs4130
```

B) Configuring Interfaces and Security Zones

Each ASA interface must be configured with a name, security level, IP address, and enabled. The recommended security levels and roles for the Birzeit Network are as follows:

- **Inside (Sec Level 100):** Represents the most trusted zone, Birzeit’s internal network.
- **DMZ (Sec Level 50):** A semi-trusted zone that hosts externally accessible servers.
- **Outside (Sec Level 0):** The untrusted zone connected to the public Internet.

To configure the **inside** interface (internal network):

```
ASA_Birzeit(config)# interface Gig1/1
ASA_Birzeit(config-if)# nameif inside
ASA_Birzeit(config-if)# security-level 100
ASA_Birzeit(config-if)# ip address 172.17.X.1 255.255.255.0
ASA_Birzeit(config-if)# no shutdown
```

Repeat the same steps to configure the **outside** and **dmz** interfaces using the appropriate interface names, security levels, and IP addresses.

C) Configuring PAT for the Internal Network

PAT enables the devices in Birzeit internal network to share a single public IP address when accessing the Internet. To map all internal devices in the 172.17.X.0/24 subnet to the ASA firewall’s **outside** interface IP address, use the following configuration:

```
ASA_Birzeit(config)# object network INTERNAL-PAT
ASA_Birzeit(config-network-object)# subnet 172.17.X.0
                                     255.255.255.0
ASA_Birzeit(config-network-object)# nat (inside,outside)
                                     dynamic interface
```

D) Configuring Static NAT for the DMZ Network

To make DMZ servers accessible from external networks, static NAT is used to map each private IP address to a public IP address from the 203.113.X.0/24 subnet. Since 203.113.X.1 is assigned to the ASA's **outside** interface and 203.113.X.2 is used by the **R. Birzeit** router, public IP assignments begin from 203.113.X.3.

To configure static NAT for the Web server (Private IP address: 172.16.X.2, Public IP address: 203.113.X.3), use the following commands:

```
ASA_Birzeit(config)# object network DMZ-WEB-NAT
ASA_Birzeit(config-network-object)# host 172.16.X.2
ASA_Birzeit(config-network-object)# nat (dmz,outside) static
                               203.113.X.3
```

Repeat these steps to configure static NAT for the **DNS** server, using the appropriate object name, private IP, and public IP addresses.

E) Enabling External Access to DMZ Servers

To allow external users to access Web and DNS services hosted in the DMZ, create an ACL that permits the required traffic from external networks to the DMZ servers.

```
ASA_Birzeit(config)# access-list OUTSIDE-ACCESS extended
                    permit tcp any host 172.16.X.2 eq www
ASA_Birzeit(config)# access-list OUTSIDE-ACCESS extended
                    permit udp any host 172.16.X.3 eq domain
```

Then, apply the OUTSIDE-ACCESS ACL to the outside interface:

```
ASA_Birzeit(config)# access-group OUTSIDE-ACCESS in interface
                    outside
```

Note:

This access list is applied **inbound** on the **outside** interface to control traffic from external users. It ensures that only specific services, such as Web and DNS, are permitted to reach DMZ servers, while preventing any traffic directed to internal network. This enhances security by restricting unnecessary or unauthorized access.

F) Enabling Return Traffic for HTTP, DNS, and ICMP to the Internal LAN

The ASA firewall is stateful, meaning it can track connections and allow return traffic for inspected protocols. This prevents legitimate responses from the DMZ or the Internet to internal clients from being blocked. By default, *DNS inspection is enabled*. To allow **HTTP responses**, you must explicitly enable HTTP inspection by updating the **global policy map** using the following commands:

```
ASA_Birzeit(config)# policy-map global_policy
ASA_Birzeit(config-pmap)# class inspection_default
ASA_Birzeit(config-pmap-c)# inspect http
```

Repeat the same steps to enable inspection for **ICMP**, allowing replies (e.g., ping responses) from the DMZ or Internet to reach the internal clients that initiated the requests.

G) Configuring Default Routing

To enable Internet access for internal and DMZ devices, configure a default route that forwards all external traffic to the **R_Birzeit** router.

```
ASA_Birzeit(config)# route outside 0.0.0.0 0.0.0.0
                        203.113.X.2
```

Note:

The destination **0.0.0.0 0.0.0.0** represents any IP address with any subnet mask. This means that any traffic not matched by a more specific route will be forwarded to the next-hop IP address 203.113.X.2 via the ASA's **outside** interface.

H) Configuring ASA as a DHCP Server for the Internal LAN

To enable dynamic IP address allocation for internal LAN devices, configure the ASA to act as a DHCP server on the **inside** interface. Since 172.17.X.1 is reserved for the **inside** interface and the printer requires a static IP configuration (e.g., 172.17.X.2) for consistent access, define a DHCP address pool of 100 addresses—sufficient for typical internal use—starting from 172.17.X.3. Also, specify the DNS server (e.g., 172.16.X.3) in the DHCP configuration.

```
ASA_Birzeit(config)# dhcpd address 172.17.X.3-172.17.X.102
                        inside
ASA_Birzeit(config)# dhcpd dns 172.16.X.3
ASA_Birzeit(config)# dhcpd enable inside
```

Repeat the same ASA configuration steps for the Gaza Education Network on **ASA_Gaza** firewall. Follow the same procedure described for **ASA_Birzeit** in Section 9.2.4 (A–H), but apply Gaza's own interface IP addresses, security levels, PAT and static NAT mappings, access lists, inspection policies, default route, and DHCP server settings.

9.2.5 Access Point Configuration

The access point provides wireless connectivity within the internal network. To configure the wireless network, go to the “**Config**” tab and set the access point with the following parameters, as shown in Figure 7:

- **Name:** BZU-WiFi
- **Security Model:** WPA2-PSK
- **Pre-Shared Key (PSK):** EnCs4130, wireless devices must enter this key to connect to the access point.
- **Encryption Type:** AES

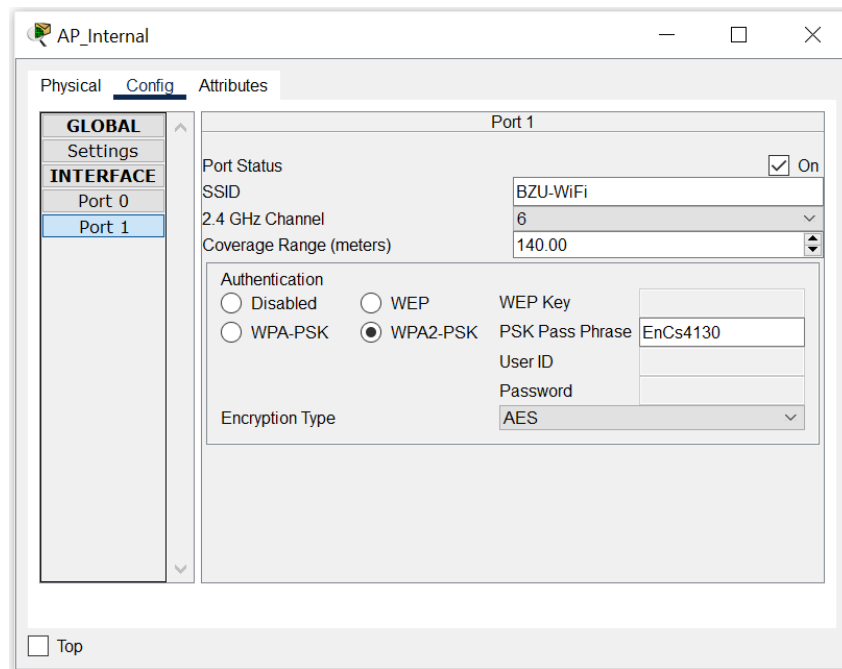


Figure 7: Configuring the access point for Birzeit internal network.

9.2.6 Connecting Wireless Devices and Configuring Dynamic IP Addresses

The tablet, laptop, and smartphone connect wirelessly to the internal network through the access point. However, unlike the tablet and smartphone, the *laptop does not have a built-in wireless interface*. To enable wireless connectivity, you must replace the laptop's wired LAN module with a **WPC300N wireless adapter** as described in **Experiment No. 3**. Figure 8 illustrates how to connect the laptop to the “BZU-WiFi” network and configure it to obtain an IP address dynamically.

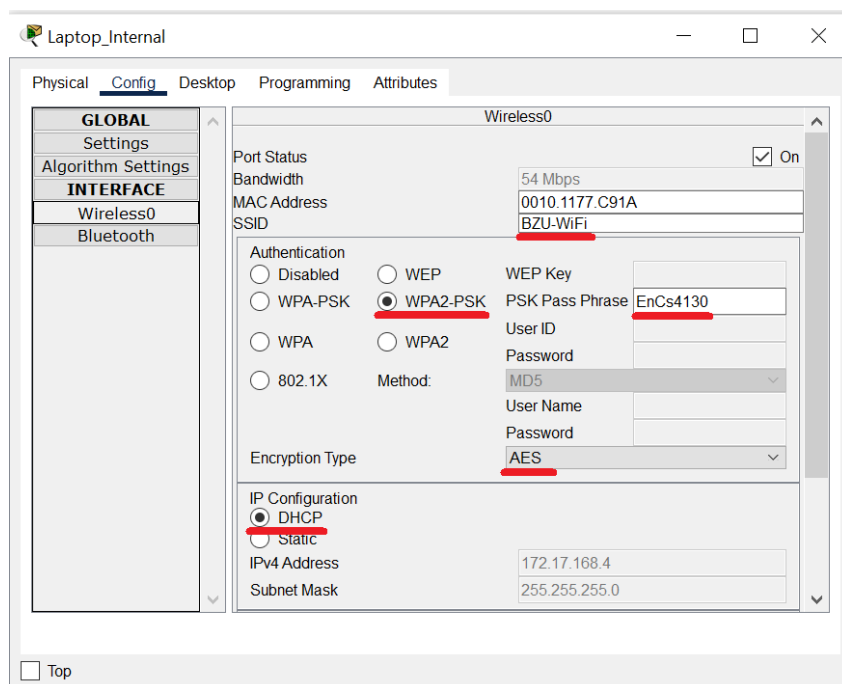


Figure 8: Connecting the laptop to BZU-WiFi and enabling dynamic IP configuration.

Repeat the same steps for the **Smartphone Internal** and **Tablet Internal** to connect them to the wireless network and obtain dynamic IP addresses. Also, configure the PCs (**PC1 Internal** and **PC2 Internal**) to use dynamic IP addressing.

9.2.7 Configuring End Devices with Static IP Addresses

The **Printer Internal** is configured with a *static IP address*, along with its *default gateway* and *DNS server* settings. Each device must be assigned a unique IP address within the valid host range of its subnet. Figure 9 shows the Printer's configuration.

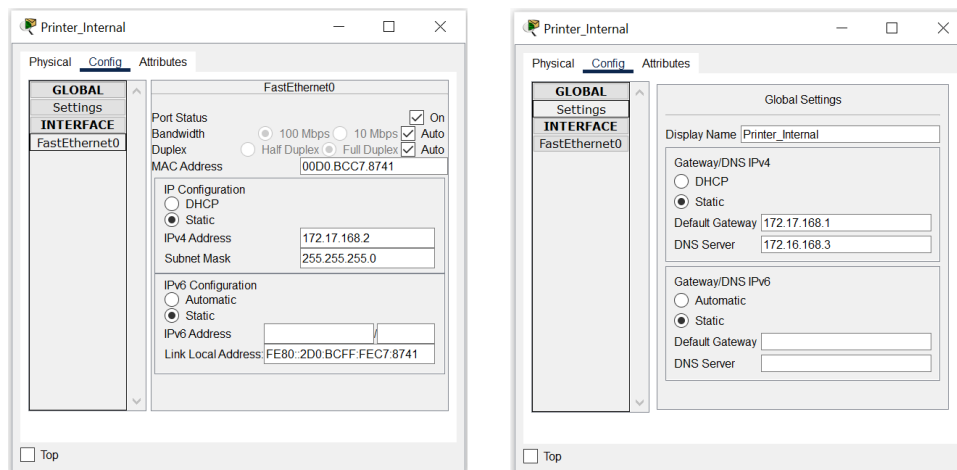


Figure 9: Configuring the Printer.

9.3 Testing Connectivity

To verify end-to-end DNS resolution as well as HTTP and ICMP connectivity between the Birzeit and Gaza Education Networks, perform the following tests:

- **Test 1 – Pinging the R_Gaza Router from Laptop_Internal:**

When pinging the **R_Gaza** router (IP: 200.1.X.1) from a device in the Birzeit Network (Internal), the ICMP communication succeeds, as shown in Figure 10. Discuss with your instructor or TA how and why this communication is successful.

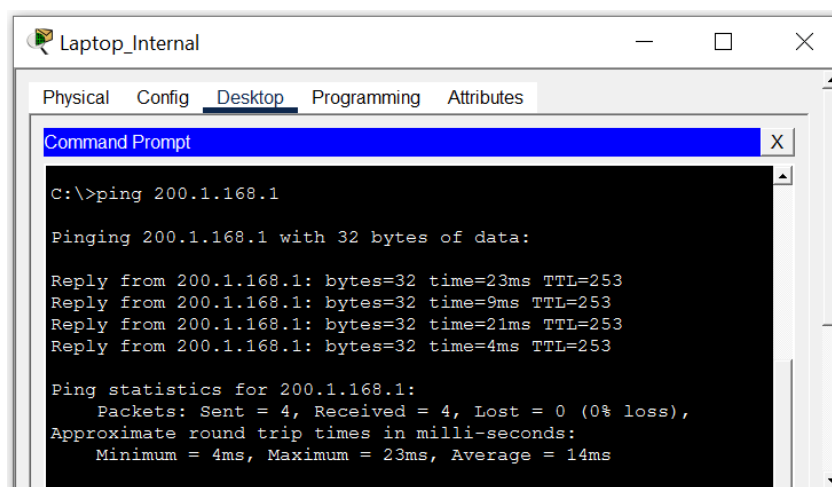


Figure 10: ICMP traffic from Birzeit Network (Internal) to the **R_Gaza** router.

- **Test 2 – Attempting to Ping a Host in Birzeit Network from PC2_Internal:**

When attempting to ping a device in the Birzeit Network from the Gaza Education Network, the traffic is blocked, as shown in Figure 11. This is because each network is protected by its own ASA firewall, which prevents direct access from an external network to internal or DMZ resources—even when ICMP inspection is enabled. These firewall policies maintain network isolation and ensure security.

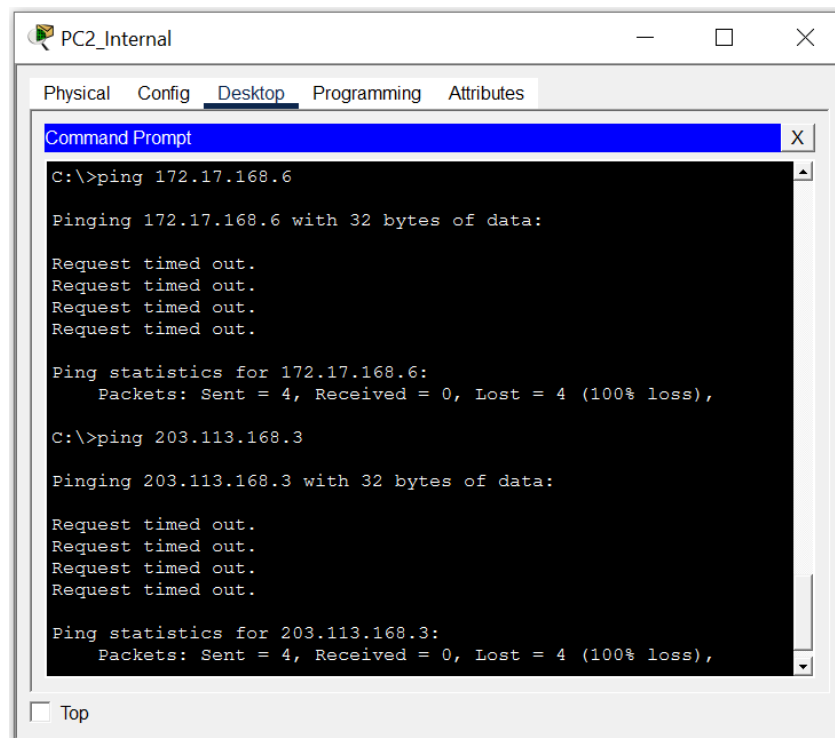


Figure 11: Blocked ICMP attempt from the Gaza Education Network (Internal) to the Birzeit Network (Internal/DMZ).

- **Test 3 – Accessing “www.gaza.com” from PC2_Internal:**

From the internal PC in the Gaza Education Network, attempt to access the Gaza DMZ web server using the domain name “www.gaza.com”, as shown in Figure 12.

Next, repeat the test using Cisco Packet Tracer’s **Simulation Mode**. Apply filters for the **DNS**, **TCP**, and **HTTP** protocols. Then, from **PC2_Internal**, attempt to access the Gaza DMZ web server again using the same domain name. Observe the sequence of packets, including:

- DNS query and reply.
- TCP three-way handshake (SYN, SYN+ACK, ACK).
- HTTP request and response.

Carefully observe the packet flow, inspect protocol exchanges, and examine how the ASA firewall processes traffic between different network zones. Analyze your observations and discuss them with your instructor or TA.

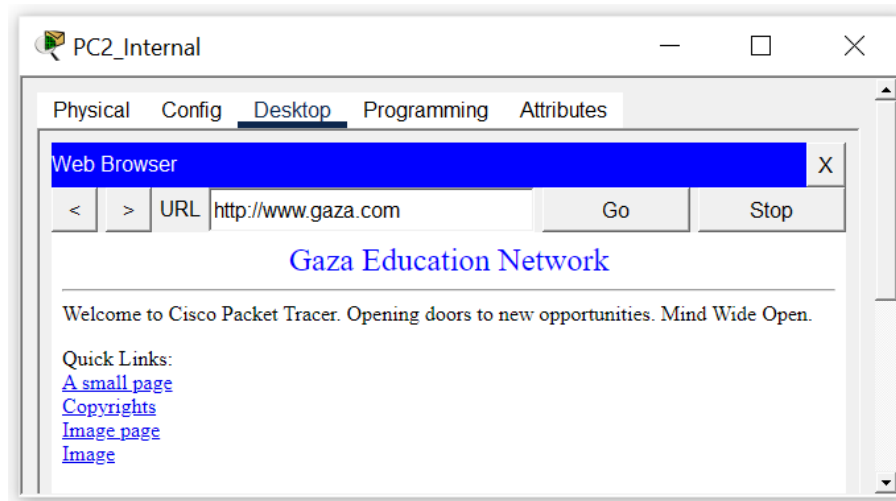


Figure 12: Accessing “www.gaza.com” from the Gaza Education Network (Internal).

- **Test 4 – Accessing “www.birzeit.edu” from PC2_Internal:**

From the internal PC in the Gaza Education Network, attempt to access the Birzeit Network’s DMZ web server using the domain name “www.birzeit.edu”, as shown in Figure 13.

Repeat the same request using **Simulation Mode**. Carefully document the packet sequence, noting the order, type, and source and destination IP addresses of each packet. Finally, analyze your findings and discuss them with your instructor or TA.

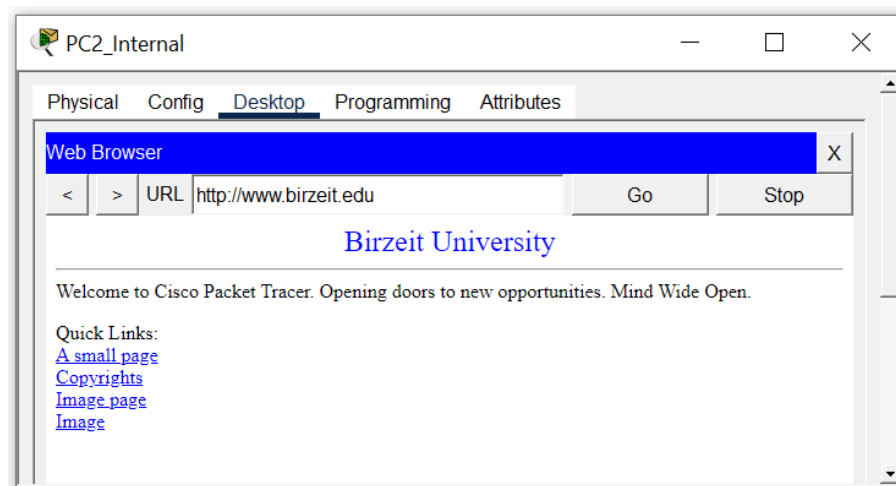


Figure 13: Accessing “www.birzeit.edu” from the Gaza Education Network (Internal).

9.4 ToDo

This section will be provided by the instructor.